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11 March 1969
SUPERSEDING
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MILITARY SPECIFICATION

COMPUTER, BALLISTICS: M13A1C

1. SCOPE

1.1 This specification covers one type of instrument known as Ballistics Computer: M13A1C, used to receive the log of range data from a range finder calibrated for range measurement in meters and computes superelevation for various ammunitions and transmits same to a gun sighting system.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture, and Inspection of
MIL-P-14232	Parts Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U. S. Army, Frankford Arsenal

F8619999	Computer, Ballistics: M13A1C

FSC 1220

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PACKAGING DATA SHEETS

8619999 Packaging of Computer, Ballistic: M13A1C

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The computer shall be manufactured in accordance with Drawing F8619999 "Ballistics, Computer: M13A1C and drawing pertaining thereto.

3.3 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Electrical.
(f) Workmanship.

3.4 Environmental.

3.4.1 Storage temperatures.- The computer shall show no evidence of physical failure and shall meet the performance requirements of 3.4.2.1 at standard ambient temperature after having been exposed and thermally stabilized at ambient temperatures of plus 160 and minus 80 degrees Fahrenheit.

3.4.2 Extreme temperature operability.- When exposed and thermally stabilized at ambient temperatures of plus 150 and minus 40 degrees Fahrenheit the computer shall meet the following requirements:

(a) Applicable requirements of 3.5.5.1 at temperatures indicated in Table IV.
(b) At minus 40 degrees Fahrenheit the outer index shall follow the inner index within one-half a graduation width over the full excursion of the range scale in a total elapsed time of 8 seconds for each direction when checked with 18 plus or minus 0.5 volts.
(c) At plus 150 degrees Fahrenheit using 28 volts, plus or minus 0.5 volts, the ammunition selector set on the AP-T, M318A1 Cam, with a load of between 14 and 18 ounce inches applied to the output shaft for the ballistics drive, and with the indexes separated by 3600 meters, the outer index shall stabilize on the inner index and shall meet the applicable

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requirements of 3.5.3 when the switch is thrown to the "ON" position. Immediately upon completion of the environmental requirements of 3.4, the interior of the computer shall be dried by clean, warm air of between approximately plus 100 degrees to 140 degrees Fahrenheit.

3.4.2.1 Post environmental check.- Each computer which has been subjected to the environmental conditions of 3.4.1 and 3.4.2 shall meet the following requirements after thermal stabilization has been reestablished at standard ambient temperature.

(a) The requirements of Table I and 3.5.4.

(b) When checked under power operation at 28 volts, plus or minus 0.5 volt, with 75 to 90 ounce inch load applied, the output shaft for the ballistics drive shall indicate the rotational values of the Table II and within the tolerances of Table III for the following ammunition and ranges:

AMMUNITION	RANGES
"AP-T, M318A1	-500, 1500 meter
HEAT, T108	-500, 4400 meter
HVAP, M332	-500, meter
HEAT, T300E54/56	-500, 1500, 4300 meter"

3.4.3 Vibration.- With the ammunition selector set on the Heat-T108 Cam, the outer index set at 3000 meters, the computer (less any shock mounts and brackets, Universal Assembly B8200862 and Output Shaft B8200801) shall be vibrated at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second for a duration of 5 minutes, plus or minus 15 seconds.

3.4.3.1 Post vibration.- Subsequent to vibration the computer shall not be damaged or show evidence of broken, loose or missing parts and it shall meet the requirements of 3.5 and 3.6.

3.4.4 Sealing.- The computer shall meet the requirements of the rain test of Specification MIL-F-13926.

3.5 Performance.

3.5.1 Output calibration.

3.5.1.1 Zero reference.- Input shaft rotation shall be measured from a precise 500 meter (zero reference) check point previously determined by the manufacturer and indicated by an index line inscribed on the input shaft in coincidence with an index on the input shaft seal cover plate.

3.5.1.2 1500 meter check point.- With the computer having a constant load of between 75 and 90 ounce inches applied to the output shaft for ballistics drive, the ammunition selector shall be set on the AP-T, M318A1 Cam. With the range correction knob set to zero and with the input shaft

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set at the 1500 meter check point, actuate the superelevation hand crank to match the outer index with the inner index within plus or minus half a line width by approaching this range first from one direction then from the opposite direction, without overtravel. Note the position of the output shaft for ballistics drive and split the backlash error at this output shaft. This is then the check point for computer output. The position of the superelevation counter shall be established from this check point and shall conform to the requirements of Table I.

TABLE I

Position of Superelevation Counter

Position of output shaft	1R-142.56° CCW	1500 Meter Check point 0°	2R-1.44° CW	6R-1.44° CW
Reading of superelevation counter in mils	3.00	9.98	20.00	40.00
Tolerance on position of output shaft	±8°	±3°	±8°	±8°

3.5.1.3 Outer index and inner index.- Using the AP-T-M18A1 Cam only, position the input shaft to correspond with each of the ranges as indicated in Table II and rotate the hand crank until the outer index is in coincidence with the inner index, at which point the rotation of this output shaft shall correspond to the value in Tables II and III. This procedure shall be repeated to check each of the eight other cams at 1500 meters only. All checks shall be made by approaching index from clockwise and counterclockwise direction.

3.5.1.4 Backlash.- Backlash of the output shaft for the superelevator shall not exceed a total of one degree at ny point of rotation of the shaft.

3.5.1.5 Outer index and range scale.- Using the cam only, rotate the hand crank until the outer index is in coincidence with the selected range on the computer range scale for each of the ranges as indicated in Table II, at which point the rotation of the output shaft for ballistics drive shall correspond to the values in Tables II and III. All checks shall be made by approaching index from clockwise and counterclockwise direction.

3.5.1.6 Servo follow-up sensitivity.- With an output shaft load of between 75 and 90 ounce inches, the ammunition selector set for the HVAP, M332 Cam, the power on at 18 plus or minus 0.5 volts and the automatic follow-up cycle completed at ranges of between 3600 and 4400 meters rotate the input shaft slowly in one direction so that it is possible to

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TABLE NO. II

Computer accuracy as determined by position of rotation of the Output Shaft based on 1500 meter reference Range for AP-T, M318A1 Cam (Dwg. #8585366).

RANGE	500 Meters	100 Meters	1500 Zero Check	2000 Meters	3000 Meters	3300 Meters	3600 Meters	4300 Meters	4400 Meters
POSITION OF INPUT SHAFT	11R 68.54° CW	4R 46.80° CW	0° Check Point	2R 334.91° CCW	7R 21.71° CCW	8R 11.23° CCW	8R 330.3° CCW	10R 261.86° CCW	10R 346.14° CCW
INPUT SHAFT INDEXING ERROR	±1°	±1°	±1°	±1°	±1°	±1°	±1°	±1°	±1°
AP-T, M318A1	1R 138.96° CCW *3.05	259.2° CCW *6.38	0° *9.98	279.36° CW *13.86	2R 192.24° CW *22.65		3R 270.72° CW *28.74		5R 226.8° CW *38.13
HE-T, T91 (2400 F/S)	1R 7.92° CCW *4.87	22.32° CW *10.29	1R 93.6° CW *16.28	2R 211.68° CW *22.92	5R 270.72° CW *38.74		8R 29.52° CW *50.39		11R 320.4° CW *69.43
HEAT T108	1R-95.8° CCW *3.65	151.2° CCW *7.88	195.12° CW *12.69	1R-257° CW *18.55	4R-253.4° CW *33.50		7R-29.5° CW *45.39		11R-62.6° CW *65.85
HEP T142E3	1R 43.2° CCW *4.38	20.16° CCW *9.70	1R 100.08° CW *16.37	2R 347.04° CW *24.80	7R 279.36° CW *48.86°	9R 248.49 CW *58.43			
HVAP, M332		1R 68.4° CCW *4.03	249.12° CCW *6.52	43.92° CCW *9.37	1R 123.8° CW *16.70		2R 198.72° CW *22.74		4R 279.36° CW *33.86
HE,M71 (REN)	1R 82.08° CCW *3.84	149.76° CCW *7.90	161.28° CW *12.22	1R 134.64 CW *16.85	3R 154.84° CW *27.20		4R 304.56 CW *34.21		6R 334.8 CW *44.63
APC-T, M82	1R 105.84° CCW *3.51	196.56° CCW *7.25	92.88° CW *11.27	1R 42.48° CW *15.57	3R 17.28° CW *25.22		4R 128.16° CW *31.76		6R 108° CW *41.48
HVAP, M304	1R 177.12 CCW	334.8° CCW	110.16° CCW	140.4° CW	2R 20.16° CW	2R 232.56° CW			
HEAT, T300E54/56	1R 228.96° CCW *1.80	1R 39.6 CCW *4.43	160.56 CCW *7.75	159.84 CW *12.20	3R 285.12 CW *28.94	5R 191.52 CW *37.64	7R 285.12° CW *48.94	15R 288.72° CW *88.99	

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count the number of times the contact closes in order to advance the output shaft by approximately 72°. Calculate the average change in superelevation per one make and break of the contacts. Repeat in the opposite direction. The average allowable superelevation increment for each range shall not exceed 10°45', and the maximum superelevation increment shall not exceed 21°30' as read at the output shaft.

3.5.2 Power operation.- With the computer having a constant load of between 75 and 90 ounce inches applied to the output shaft, the rotation of the output shaft shall correspond to the values indicated in Tables II and III for the ammunition and ranges specified therein. The test shall be run by allowing the outer index to approach the inner index from both increasing and decreasing range with power on continuously at 28 plus or minus 0.5 volts. The computer shall also meet the accuracy requirement of Tables II and III for the following ammunition and at ranges indicated when checked at 18 plus or minus 0.5 volts.

AMMUNITION AND RANGES FOR 18 VOLT CHECK

HE, M71	1500, 3000 and 4400 meters
HEAT, T300E54/56	1500, 2000 and 4300 meters
HEAT, T108	1500, 3600 and 4400 meters

All checks shall be made by approaching index from both clockwise and counterclockwise direction.

3.5.3 Output shaft tolerance.- When the output shaft rotation of the computer, as read in degrees is equivalent to the superelevation value in mils as shown in column "A", the output shaft tolerance shall be within the corresponding value as shown in column "B" of Table III.

TABLE III

OUTPUT SHAFT TOLERANCE

A Output shaft rotation (Superelevation in mils)	B Degrees	C Mils*
0 up to but less than 6	± 7.20	± 0.10
6 up to but less than 11	± 10.80	± 0.15
11 up to but less than 26	± 14.40	± 0.20
26 up to but less than 31	± 18.00	± 0.25
31 up to but less than 36	± 25.20	± 0.35
36 up to but less than 41	± 28.80	± 0.40
41 up to but less than 61	± 36.00	± 0.50
61 up to but less than 86	± 43.20	± 0.60
86 up to but less than 98.5	± 50.40	± 0.70

* The data contained in column "C" is informational only.

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3.5.4 Stability.- Using 28 plus or minus 0.5 volts with an output shaft load of between 14 and 18 ounce inches, range correction knob set to introduce minus 15 percent within one percent, the rotation of the input shaft between minimum range and 1500 meters shall not cause the computer servo to display any self-sustaining or oscillatory operation when checked on any of the 6 cams within the computer. This is not required for the spare cams. A maximum of 8 changes in direction of the servo motor, as a settling out action, shall be considered as satisfactory operation. These 8 changes may be counted as clicks.

3.5.5 Component operation.

3.5.5.1 Operational values.- The operational values shall not exceed the values given in Table IV at temperatures indicated.

TABLE IV

OPERATIONAL VALUES

Control	Normal Room Temperature up to 150°F	Cold Temperature (Minus 40°F)	Operation
Input Shaft	16 oz. in.	45 oz. in.	Rotation
Range Correction Knob	34 oz. in.	45 oz. in.	Rotation
Range Correction Knob	20 lbs.	24 lbs.	Pull out force
Ammo Selector	60 lb. in.	80 lb. in.	To lift follower
Hand Crank	6 lb. in.	20 lb. in.	To rotate while engaged
Hand Crank	12 lbs.	20 lbs.	To hold in engagement
Interrupter Switches --------	Mechanically & Electrically Operable	Mechanically & Electrically Operable	In accordance with Par. 3.6.3
Reset Button	15 lbs.	40 lbs.	In accordance with Par. 3.6.3

3.5.5.2 Time of response.- At standard ambient temperature (plus 60 to 90 degrees Fahrenheit), with the output shaft load as specified in paragraph 3.5.2 applied, the inner and outer indexes positioned as specified in Table V, and the ammo selector set on the HE, M71 Cam, the outer index shall stabilize on the inner index within the elapsed time specified, after power has been applied.

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TABLE V

TIME OF RESPONSE

Position of Outer Index	Position of Inner Index	Voltage (plus or minus 0.5.V.)	Total Elapsed Time
4400 meters	500 meters	28 V.	4 seconds
500 meters	4400 meters	28 V.	4 seconds
500 meters	4400 meters	18 V.	8 seconds
4400 meters	500 meters	18 V.	8 seconds

3.5.5.3 Circuit breaker.

3.5.5.3.1 Breaker open.- With the motor at standard temperature, with 28 plus or minus 0.5 volts applied, the two indexes separated by at least 200 meters, the output shaft for the ballistics drive locked (immovable), the circuit breaker shall open in 90 seconds or less.

3.5.5.3.2 Breaker closed.- At standard ambient with the load as specified in paragraph 3.5.2 applied to the output shaft for the ballistics drive, the computer AP-T-,M31841 Cam, the input shaft shall be oscillated for a period of 45 seconds over a spread of 1000 meters, plus or minus 100 meters between ranges of 1200 to 2800 meters, at a rate of 4 to 4.5 revolutions per second, during which time the circuit breaker shall not open.

3.5.5.4 Hand crank and input shaft travel.

3.5.5.4.1 Hand crank.- The rotation of the superelevation hand crank within the limits of its stops shall cover the excursion from minus 1.5 mils, plus or minus 0.1 mil, to plus 98.5 mils, plus or minus 0.5 mil, as indicated on the superelevation counter when the range correction knob is set at zero.

3.5.5.4.2 Input shaft.- The range input shaft shall rotate through an excursion of 24.0 revolutions plus 0.1 revolution and shall cover the range from 480 to 4575 meters within this excursion, as measured from the 1500 meter check point of the input shaft.

3.5.6 Superelevation cams.- The three cams, HE-T, T91; HVAP, M304 and HEP, T142E3, shall be placed in the computer and checked, after which these same three cams shall be removed from the computer and placed in the spare cam box. Six cams shall then be placed in the computer in the following sequence starting with the AP-T, M318A1 and following with APC-T, M82 - HEAT, T108 - HVAP, M332 - HE, M71 and HEAT, T300E54/56. The computer shall be shipped with the same nine cams with which it qualified.

3.6 Electrical check.

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3.6.1 Insulation breakdown.- Insulation breakdown shall be in accordance with Specification MIL-F-13926 except that breakdown voltage of 500 volts RMS shall be substituted for 750 volts RMS.

3.6.2 Insulation resistance.- Insulation resistance shall be in accordance with Specification MIL-F-13926 except that the resistance value shall not be less than 2 megohms and a dc potential of 500 volts shall be used in place of 800 volts. All lamps shall be removed from sockets prior to tests.

3.6.3 Continuity.- The electrical circuits shall be checked for continuity, and shall conform to the wiring diagram. When the hand crank is engaged the interrupter switch shall open the motor circuit and energize the pilot light. Pressing of the reset button shall cause the motor circuit to be closed and the pilot light to be extinguished. Actuation of the ammo selector handle shall cause the ammunition selector interrupter switch to open the motor circuit. Resetting of the handle shall cause switch to complete the circuit.

3.6.4 Illumination.- The internal lighting device shall provide adequate illumination so that the inner and outer indexes of the range dial and the ammunition index shall be clearly defined and readable when observed in a darkened inclosure.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 computers plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that

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it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein, shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105. The sampling plans specified in Tables VI, VII & VIII shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE VI - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 1.0% Defective			
101.	Vibration	3.4.3, 3.4.3.1	4.6.1.5, 4.6.1.5.1
102.	Zero reference	3.5.1.1	4.6.2.1
103.	1500 meter check point	3.5.1.2	4.6.2.3
104.	Outer index and inner index	3.5.1.3	4.6.2.4
105.	Backlash	3.5.1.4	4.6.2.5
106.	Outer index and range scale	3.5.1.5	4.6.2.6
107.	Servo follow-up sensitivity	3.5.1.6	4.6.2.7
108.	Power operation	3.5.2	4.6.2.8
109.	Output shaft tolerance	3.5.3	4.6.2.9
110.	Stability	3.5.4	4.6.2.10
111.	Time of response	3.5.5.2	4.6.3.2
112.	Circuit breaker open	3.5.5.3.1	4.6.3.3.1

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CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
113. Circuit breaker closed	3.5.5.3.2	4.6.3.3.2
114. Hand crank travel	3.5.5.4.1	4.6.3.4.1
115. Input shaft travel	3.5.5.4.2	4.6.3.4.2
116. Superelevation cams	3.5.6	4.6.3.5
117. Insulation breakdown	3.6.1	4.6.3.6.1
118. Insulation resistance	3.6.2	4.6.3.6.2
119. Continuity	3.6.3	4.6.3.6.3
120. Illumination	3.6.4	4.6.3.6.4

MINOR: NONE

NOTE: The tests in Table VI shall be conducted at a temperature between +60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One computer shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table VII.

TABLE VII

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Materials	3.1	Applicable drawings - material and general specifications
302. Fabrication	3.2	Applicable drawings - visual
303. General specification	3.3	MIL-F-13926 - visual

4.4.2 Environmental.- Three computers shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table VIII and shall then meet the requirements and tests in Table I.

TABLE VIII

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304. Storage temperature (+160° and -80°F)	3.4.1	4.6.1.1
305. Operability temperature (+150° and -40°F)	3.4.2, 3.5.5.1	4.6.1.2, 4.6.3.1

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CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
306. Post environmental check	3.4.2.1	4.6.1.3
305. Sealing	3.4.4	4.6.1.4

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents (IEL) when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Storage temperatures.- Testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.1. The computer shall be exposed to each specified temperature extreme for a minimum of 5 hours. Subsequent to the temperature exposures the computer shall be stabilized in an ambient temperature between +60° and +90°F and subjected to tactile

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and visual examination to determine compliance with the requirements of 3.4.1. Caution must be exercised during this test to avoid subjecting the computer to thermal shock.

4.6.1.2 Extreme operability temperatures.- These tests shall be conducted in accordance with the temperature conditions in 3.4.2. When thermally stabilized at each of the extreme temperatures specified, the computer shall be tested in accordance with the following:

(a) The controls listed in Table IV shall not exceed the applicable torque and force values at the temperatures specified therein for the operational tests of those controls to determine conformance with 3.4.2a. The equipment required for these tests is listed on Drawings F7659613, F7659614 and F8586721. The reset button shall be inspected using standard measuring equipment capable of measuring within one-quarter (1/4) pound.

(b) Rotate the input shaft from the clockwise stop to the counterclockwise stop; observe the outer index; it shall follow the inner index in the required time limit at the specified voltage in conformance with 3.4.2b. Repeat the test by rotating the input shaft in the opposite direction. The circuit breaker switch shall be in the "ON" position. This test shall be performed at minus 40 degrees Fahrenheit.

(c) Set the ammunition selector on the cam specified, the range correction knob to zero, and apply the specified torque load to the computer output shaft. Position the computer inner index at 500 meters and the outer index at 4400 meters on the range scale. When the circuit breaker switch is thrown to the "ON" position, the outer index shall "settle out" (cease oscillatory operation) coincident to the inner index within eight (8) changes (clicks) in direction of the computer servo in compliance with 3.4.2c and 3.5.3. The computer shall again meet this requirement when the test is repeated with the outer index positioned at 500 meters and the inner index positioned at 4400 meters on the range scale.

4.6.1.3 Post environmental temperature checks.- Following temperature environmental tests, the computer shall be thermally stabilized at room temperature (60° to 90°F) and tested for the following to determine comliance with 3.4.2.1:

(a) The tests of 4.6.2.3 and 4.6.2.9

(b) With the required load applied to the output shaft and at the specified voltage, the computer shall be tested in accordance with 4.6.2.8.

4.6.1.4 Sealing (water resistance).- This test shall be conducted in accordance with Procedure I of Specification MIL-F-13926 to determine compliance with 3.4.4. The computer shall show no evidence of water penetration as a result of this test. The receptacle shall be covered with waterproof tape during this test. The computers that are opened for environmental drying and examination for water penetration shall be resealed to meet the requirements of the drawings prior to submission for final acceptance.

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4.6.1.5 Vibration.- Each computer in the sample shall be mounted on Fixture F7560085 using Adapter F7659608 and vibrated in accordance with the requirements specified in 3.4.3.

4.6.1.5.1 Post vibration, physical failure.- Following vibration, the computer shall be subjected to a visual and tactile examination for damage such as loose, bent or broken parts, etc. to determine compliance with 3.4.3.1. The computer shall then be subjected to all the following tests of this specification.

4.6.2 Output calibration.

4.6.2.1 Zero reference.- The 1500 meter check point as determined previously by the manufacturer shall be visually checked for compliance with 3.5.1.1. It shall be indicated by an index line inscribed on the input shaft in coincidence with an index on the input shaft seal plate cover.

4.6.2.2 The following tests shall be conducted using Test Fixture A8270832 and in accordance with the operating instructions specified on Drawing F8565296 using appropriate metric scales. The computer input shaft and the fixture input mechanism shall be oriented to the 1500 meter check point.

4.6.2.3 1500 meter check point.- Set the ammunition selector on the cam specified and the range correction knob on zero. When the mil value specified in Table I is indicated on the computer superelevation mil counter, the rotational position of the output shaft for the ballistics drive shall not exceed the tolerance specified in Table I for compliance with 3.5.1.2.

4.6.2.4 Outer index and inner index.- Set the ammunition selector on the cam specified and the range correction knob on zero. When the rotational position of the computer input shaft corresponds to the value for each of the ranges indicated in Table II, and the computer outer and inner indexes are coincident, the rotational position of the ballistics drive output shaft shall not exceed the tolerances as specified in Table III. This test shall be repeated to check each cam at 1500 meters only for compliance with 3.5.1.3.

4.6.2.5 Backlash.- Backlash of the computer superelevator output shaft shall not exceed the tolerance specified in 3.5.1.4 when measured at four (4) points of shaft rotation. The four (4) points of measure shall be approximately ninety degrees (90°) apart.

4.6.2.6 Outer index and range scale.- Set the ammunition selector on the cam specified and the range correction knob on zero. When the computer outer index is coincident to the selector range mark on the computer range scale, the rotational position of the output shaft for the ballistics drive shall correspond to the values of Table II and not exceed the tolerances of Table III for compliance with 3.5.1.5.

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4.6.2.7 Servo follow-up sensitivity.- Set the ammunition selector on the cam specified, the range correction knob on zero, and the inner and outer index to a range between 3600 meters and 4400 meters. With the computer in power operation, the average and maximum recorded increment change (closing of computer servo contacts) resulting from a total rotation of the output shaft 72° clockwise and 72° counterclockwise shall not exceed the tolerance specified in 3.5.1.6. Superelevation increment changes shall be measured at the computer output shaft for the ballistics drive.

4.6.2.8 Power operation.- This test shall be conducted as specified on Drawing F8565296 with the range correction knob set to zero. The tests in 4.6.8.1 and 4.6.8.2 shall be conducted to determine compliance with 3.5.2.

4.6.2.8.1 Input at 28 volts.- With the input voltage set at 28 plus or minus 0.5 volts and the circuit breaker switch in the "ON" position, the accuracy of each cam shall be tested at the range listed in Table II for computer output shaft rotation. When the input shaft is rotated to a value indicated in Table II for a given range, the rotational position of the output shaft shall correspond to the value for that range in Table II within the tolerances specified in Table III to determine compliance with 3.5.2.

4.6.2.8.2 Input at 18 volts.- With the input voltage set at 18 plus or minus 0.5 volts, the computer shall meet the accuracy requirements of Tables II and III for the cams and ranges specified to determine compliance with 3.5.2.

4.6.2.9 Output shaft tolerance.- Rotation of the output shaft shall be determined by representative mils in superelevation (see Column A of Table III) and shaft rotation measurement in degrees shall be within the tolerance specified in Column C of Table III for conformance with 3.5.3.

4.6.2.10 Stability.- Prepare the computer for this test in accordance with Drawing F8565296; set the range correction knob to minus fifteen percent (-15%) within one percent (1%), and adjust the input voltage to 28 plus or minus 0.5 volts. Rotate the computer index from minimum range to 1500 meters; throw the circuit breaker switch to "ON" position to energize the servo motor. The outer index shall "settle out" (cease oscillatory operations) coincident to the inner index within eight (8) changes (clicks) in direction of the computer servo. The inner index shall then be rotated to four (4) different ranges between 1500 meters and minimum range, and the outer index again shall "settle out" coincident to the inner index within eight (8) changes in direction of the servo. Set the range correction knob to plus fifteen percent (+15%) within one percent (1%), and repeat the aforementioned test. All cams installed in or accompanying the computer shall be subjected to this test to determine compliance with 3.5.4.

4.6.3 Component operation.

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4.6.3.1 Operational values.- Operating tests shall not exceed the values listed in Table IV at the temperatures indicated to determine compliance with 3.5.5.1

4.6.3.2 Time of response.- Set the ammunition selector on the cam specified and the range correction knob on zero. When the circuit breaker switch is thrown to the "ON" position, and the servo motor is energized with the load in 3.5.2 applied to the output shaft, the computer outer index shall stabilize on the inner index within the time cycle specified in Table V for compliance with 3.5.5.2.

4.6.3.3 Circuit breakers.

4.6.3.3.1 Circuit breaker, open.- The inner and outer index shall be separated by the specified distance, and the output shaft for the ballistics drive immobilized. When the circuit breaker switch is thrown to the "ON" position, and the servo motor is energized, the circuit breaker shall open (switch move to "OFF" position) within the time cycle specified in 3.5.5.3.1.

4.6.3.3.2 Circuit breaker, closed.- Set the ammunition selector on the cam specified, the range correction knob on zero, and the inner and outer index coincident on 2000 meters of range. The computer input shaft shall be oscillated as specified in 3.5.5.3.2 with the circuit breaker switch in the "ON" position and the servo motor energized. The circuit breaker shall not open (switch move to "OFF" position) as a result of this oscillation.

4.6.3.4 Hand crank and input shaft travel.

4.6.3.4.1 Hand crank travel.- The test for hand crank travel shall be conducted by manually rotating the hand crank and observing the superelevation counter to determine the excursion specified in 3.5.5.4.1.

4.6.3.4.2 Input shaft travel.- The test for input shaft travel shall be conducted by manually rotating the range input shaft through the excursion specified in revolutions in 3.5.5.4.2 to determine if full range coverage (in meters) is in compliance with 3.5.5.4.2 as measured from the 1500 meter checkpoint of the input shaft.

4.6.3.5 Superelevation cams.- The inspection sequence and arrangement of cams in the computer shall be as specified in 3.5.6 by means of visual examination.

4.6.3.6 Electrical checks.- The following tests shall be conducted at any time subsequent to vibration. Before testing, remove lamp bulbs and disconnect capacitor from computer circuit.

4.6.3.6.1 Insulation breakdown.- Procedure I in this test shall be conducted in accordance with Specification MIL-F-13926 for insulation breakdown to meet the requirements of 3.6.1.

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4.6.3.6.2 Insulation resistance.- This test shall be conducted in accordance with Specification MIL-F-13926 for insulation resistance, except the values in 3.6.2 shall apply.

4.6.3.6.3 Continuity.- A continuity test of the computer circuits shall be made using standard measuring equipment to determine conformance with the wiring diagram. Actuation of the hand crank, reset button, and ammunition selector handle shall cause the computer to function.

4.6.3.6.4 Illumination.- The illuminated inner and outer indexes of the range dial and ammunition index of the computer shall meet the requirements when observed in a darkened enclosure with the unaided eye.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with the applicable packaging data sheet listed in Section 2. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Ballistics Computer: M13A1C is an electromechancial component of the primary fire control system for 90 MM Gun Tank: M48A2C. The ballistics computer manually or electrically determines and relays to the sighting system the amount of gun superelevation required for specific target range, and ammunition type.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of preservation, packaging, and packing.
c. Applicable stock number.
d. Applicable packaging data sheet (see 5.1).
e. Requirement for first article inspection.

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